

# YASH BAVADIYA

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## EDUCATION

Indian Institute of Technology, Jodhpur

Aug 2024 – Jul 2028

Bachelor's Degree - Computer Science and Engineering - CGPA - 7.21

Jodhpur, Rajasthan

## TECHNICAL SKILLS

**Languages:** C++, C, Python, TypeScript, JavaScript, SQL

**Core CS:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, System Design, Distributed Systems

**Technologies/Frameworks:** Qt (C++), Node.js, Express, React, Next.js, Prisma, PostgreSQL, MySQL, MongoDB, Redis, WebSockets, Docker, Linux/UNIX, REST APIs, TCP/IP

**Tools:** Git, GitHub Actions, Nginx, Postman, Vercel, VS Code

## EXPERIENCE

Google Summer of Code 2026

May 2026 – Aug 2026

Kdenlive @KDE Community | Open Source Contributor, C++ and Qt

Remote

- Selected for **GSoC 2026** among 1,200 global contributors to design and implement Kdenlive's **RGB curve, gradient, and speed ramp editors** in **C++ and Qt** within a 500k+ line production codebase.
- Shipped **20 contributions** across Kdenlive and upstream repositories, resolving an **11-year-old bug** with a gap-detection algorithm now running for **41,000 monthly installs**, reviewed and merged by core maintainers.

Undergraduate Researcher

Jul 2025 – Nov 2025

Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder

IIT Jodhpur

- Engineered an offline **Retrieval-Augmented Generation pipeline** integrating FAISS vector search with 4-bit quantized Phi-3 and Flan-T5-XL, achieving a **+71.77% average improvement** across 8 ROUGE metrics.
- Attained **3 second query latency** on 300-page PDFs and deployed the solution to **15 IIT Jodhpur students** during semester exams across 3 feedback-driven iterations.

## PROJECTS

Flock | [GitHub](#) | [npm](#) | TypeScript, WebSocket, Redis, React, Docker

Feb 2026 – Jun 2026

- Released 3 npm packages enabling multiplayer cursors and live presence in web applications via a **10-line integration**, installable as a **55MB** self-hosted Docker image or npx.
- Implemented Redis pub/sub message relay across server instances with keyspace-notification-driven user eviction, reducing stale presence state to under **30 seconds** on ungraceful disconnects.
- Delivered cursor position smoothing with continuous mid-animation target updates, exponential backoff reconnection, and **58 tests** covering throttling, time-to-live eviction, and message delivery.

DaemonDoc | [GitHub](#) | [Live](#) | TypeScript, Express, MongoDB, Redis (BullMQ), Docker

Jan 2026 – May 2026

- Automated documentation generation on git push via GitHub Webhooks on a **Redis (BullMQ)** job queue, sustaining an **85% success rate** across **959 production runs** over 68 repositories for **50 users**.
- Cut per-run API cost roughly **10x** with a two-stage router sending file paths and token estimates, never file content, to a lightweight model to select context, with failover across 6 provider keys.
- Engineered content-addressed section hashing over a parsed document tree, with idempotency and loop guards on the webhook path and validation that skips a commit rather than corrupt a user's file.

IIT Jodhpur PHC System | [GitHub](#) | Node.js, Express, PostgreSQL, Prisma, Docker

Jan 2026 – Apr 2026

- Spearheaded backend architecture to replace a paper-based workflow with a permission-based platform featuring **6 roles, 63 endpoints, and 20 database models** using JSON Web Token and LDAP authentication.
- Verified reliability via **48 test cases** and load tests at **20 concurrent users**, securing production deployment approval projecting **5 staff-hours saved per day**.

## COURSEWORK

- |                                  |                               |                              |
|----------------------------------|-------------------------------|------------------------------|
| • Data Structures and Algorithms | • Database Management Systems | • Discrete Mathematics       |
| • Object-Oriented Programming    | • Software Engineering        | • Pattern Recognition        |
| • Operating Systems              | • Machine Learning            | • Probability and Statistics |

## ACHIEVEMENTS

- Solved **450+** algorithmic problems on [LeetCode](#) in C++; Max Rating **1429** on Codeforces across rated contests.
- JEE Advanced 2024: AIR **3449** of 180,000+ candidates; JEE Main 2024: **99.65** percentile, AIR **5680**.
- Merged **84+ open-source contributions** across GitHub and GitLab, shipping features and bug fixes to [Kdenlive](#), [Vitest](#), [Wagtail](#), and [croc](#).

## LEADERSHIP

- Junior Core Team Member** at DevUp Labs, IIT Jodhpur; mentored 30 students and facilitated 2 open-source workshops.
- Core Member** at E-Cell, IIT Jodhpur; supported 3 entrepreneurship events and initiatives over 9 months.
- Core Member** at Programming Club, IIT Jodhpur; co-organized 5 contests and 10 sessions on competitive programming.